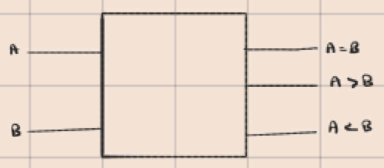
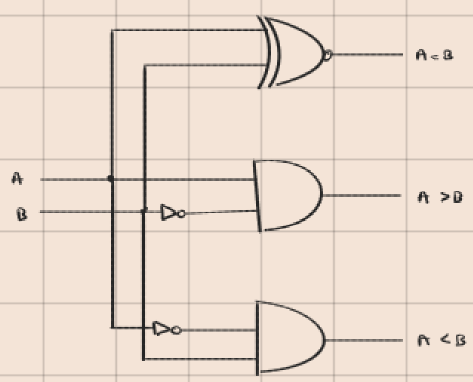
Comparator



1 - bit Comparator

A	B	A=B	A>B	A<B
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0



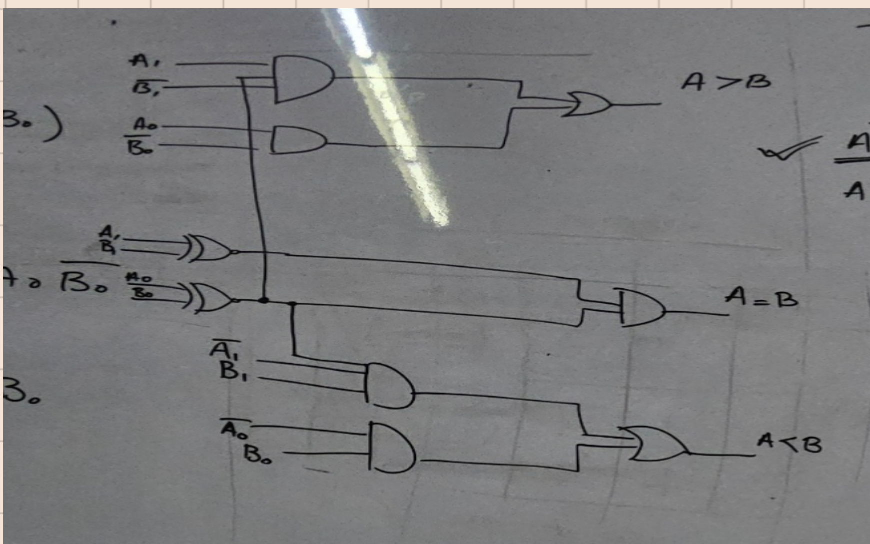
2-bit Comparator

A_1	A_0	B_1	B_0	A=B	A>B	A<B
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

$A = B \;=\; (A_1 \oplus B_1)(A_0 \oplus B_0)$

$A > B \;=\; A_1\overline{B_1}(A_0 \oplus B_0) + A_0\overline{B_0}$

$A < B \;=\; \overline{A_1}\,\overline{B_1}(A_0 \oplus B_0) + \overline{A_0}B_0$



4 bit

A_3	A_2	A_1	A_0	B_3	B_2	B_1	B_0	$A>B$	$A<B$	$A=B$
0	0	0	0	0	0	0	0			
$\downarrow$ 128	$\downarrow$ 64	$\downarrow$ 32	$\downarrow$ 16	0	0	0	1			
				0	0	1	0			
1				0	0	1	1			
$\downarrow$ 128				0						
				0						
				0						
				0						